

Hover Lift Estimates for a Two-Blade VTOL Nacelle Rotor

Jon Farhat (inputs)

January 23, 2026

1 Inputs (Per Nacelle)

- Blades: $N_b = 2$, variable pitch ($0\text{--}10^\circ$)
- Chord: $c = 0.030\text{ m}$
- Electrical power in hover: $P_e = 400\text{ W}$ (about 27 A on 4S)
- Nacelle mass: $m_n = 1.292\text{ kg}$, gravity $g = 9.81\text{ m/s}^2$
- Air density: $\rho = 1.225\text{ kg/m}^3$
- Radii evaluated: $R = 0.435\text{ m}$ and $R = 0.600\text{ m}$

2 Geometry

Disk area:

$$A = \pi R^2 \tag{1}$$

Solidity (shows where chord enters blade-element models):

$$\sigma = \frac{N_b c}{\pi R} \tag{2}$$

Computed:

$$A_{435} = \pi(0.435)^2 = 0.594\text{ m}^2, \quad \sigma_{435} = \frac{2(0.030)}{\pi(0.435)} = 0.0439 \tag{3}$$

$$A_{600} = \pi(0.600)^2 = 1.131\text{ m}^2, \quad \sigma_{600} = \frac{2(0.030)}{\pi(0.600)} = 0.0318 \tag{4}$$

3 Baseline Weight (Nacelle Only)

$$W_n = m_n g = 1.292 \times 9.81 = 12.67\text{ N} \tag{5}$$

4 Hover Thrust From Power (Momentum Theory)

Induced velocity in hover:

$$v_i = \sqrt{\frac{T}{2\rho A}} \quad (6)$$

Ideal induced power:

$$P_{i,\text{ideal}} = T v_i = \frac{T^{3/2}}{\sqrt{2\rho A}} \quad (7)$$

Solve for thrust vs *shaft* power P_s :

$$T_{\text{ideal}}(P_s) = \left(P_s \sqrt{2\rho A}\right)^{2/3} \quad (8)$$

We model electrical-to-effective power with an overall effectiveness η :

$$P_s \approx \eta P_e \quad \Rightarrow \quad T_{\text{real}} \approx \eta^{2/3} T_{\text{ideal}}(P_e) \quad (9)$$

5 Case Results (Per Nacelle)

Using $P_e = 400$ W:

Case A: $R = 0.435$ m

$$T_{\text{ideal},435} = \left(400 \sqrt{2(1.225)(0.594)}\right)^{2/3} = 61.5 \text{ N} \quad (10)$$

$$\frac{T_{\text{ideal},435}}{g} = 6.27 \text{ kgf} \quad (11)$$

$$m_{\text{payload,ideal},435} = 6.27 - 1.292 = 4.98 \text{ kg} \quad (12)$$

Case B: $R = 0.600$ m

$$T_{\text{ideal},600} = \left(400 \sqrt{2(1.225)(1.131)}\right)^{2/3} = 76.2 \text{ N} \quad (13)$$

$$\frac{T_{\text{ideal},600}}{g} = 7.77 \text{ kgf} \quad (14)$$

$$m_{\text{payload,ideal},600} = 7.77 - 1.292 = 6.48 \text{ kg} \quad (15)$$

6 Realistic Bracket Using η

Pick example effectiveness values:

$$\eta \in \{0.35, 0.50, 0.60\} \quad \Rightarrow \quad \eta^{2/3} \in \{0.497, 0.630, 0.711\}.$$

Table 1: Hover thrust and payload above nacelle mass (per nacelle) at $P_e = 400$ W.

Case	η	Thrust (kgf)	Payload above nacelle (kg)
$R = 0.435$ m	Ideal ($\eta = 1$)	6.27	4.98
$R = 0.435$ m	0.35	3.11	1.82
$R = 0.435$ m	0.50	3.95	2.66
$R = 0.435$ m	0.60	4.46	3.17
$R = 0.600$ m	Ideal ($\eta = 1$)	7.77	6.48
$R = 0.600$ m	0.35	3.86	2.57
$R = 0.600$ m	0.50	4.90	3.60
$R = 0.600$ m	0.60	5.53	4.24

7 Two Nacelles (Total Payload Above Both Nacelles)

If you have two identical nacelles (two rotors total), approximate total payload above both nacelles:

$$m_{\text{payload,total}} \approx 2 m_{\text{payload,per nacelle}}$$

(assuming no large rotor-rotor interference and comparable operating points).